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# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**
**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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**SDC HACKATHON 2026**

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | TriHack |
| **Project / Solution Name** | SmartSME Procurement |
| **Name (Leader)** | Ng Xin Rou |
| **Email & Phone (Leader)** | ERA2509054@xmu.edu.my, +6017-2724083 |
| **University Affiliation(s)** | Xiamen University Malaysia |
| **Live Prototype URL** | https://tri-hack-sdc-hackaton.replit.app |
| **Code Repository URL (public / access granted to judges)** | https://github.com/ChLing26/SDC_Hackathon_Trihack |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

In this architecture hosted within a cloud environment, the React browser UI communicates with the Express API via same-origin or api requests using session cookies for authentication. The Express API acts as the central orchestrator, directly querying a PostgreSQL database that manages accounts, requests, approvals, invoices, and signatures. To handle AI capabilities, the Express API routes requests through a private Python advice service connected to the Gemini API for advisory features, while separately utilizing an OpenAI integration to generate recommendations derived from historical approved data records.

### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| **TypeScript** 5.9.3 | Type-checks the TypeScript application and libraries. | Apache 2.0 |
| **Node.js** 24.x | Runs the TypeScript/JavaScript services and build tools. The project specifies the major version, not a minor or patch. | MIT |
| **Python** 3.12.x | Runs the private supplier-advice service. The project specifies the major/minor version. | PSF License |
| **React / React DOM** 19.1.0 | Builds and renders the ProcureFlow interface. | MIT |
| **Vite** 7.3.6 | Develops and builds the web applications. | MIT |

| Name & Version | Purpose in Project | License |
|---|---|---|
| **Tailwind CSS** 4.3.3 | Provides utility-based styling. | MIT |
| **Wouter** 3.13.0 | Handles client-side page routing. | MIT |
| **Radix UI** Dialog 1.1.23 and Accordion 1.2.20 | Provides accessible interface primitives. | MIT |
| **Zod** 3.25.76 | Validates API data in TypeScript. | MIT |
| **Lucide React** 0.545.0 | Supplies UI icons. | ISC |
| **Recharts** 2.15.4 | Supplies chart components. | MIT |
| **jsPDF / html-to-image** 4.2.1 / 1.11.13 | Creates invoice PDFs from browser-rendered content. | MIT |
| **Express** 5.2.1 | Serves the API and handles application routes. | MIT |
| **Drizzle ORM** 0.45.2 | Connects the TypeScript backend to PostgreSQL. | Apache 2.0 |
| **pg (node-postgres)** 8.23.0 | PostgreSQL driver used by the backend. | MIT |
| **Pino / pino-http** 9.14.0 / 10.5.0 | Server-side structured logging. | MIT |
| **OpenAI SDK** 6.49.0 | Calls OpenAI for recommendation summaries. | Apache 2.0 |
| **FastAPI / Uvicorn** 0.161.1 / 0.54.0 | Defines and runs the private Python API. | MIT / BSD 3-Clause |
| **Pydantic** 2.14.1 | Validates Python API requests and data. | MIT |
| **Google Gen AI SDK** 2.33.0 | Connects the Python advice service to Gemini. | Apache 2.0 |
| **Requests / HTTPX** 2.34.2 / 0.28.1 | Python HTTP client libraries. | Apache 2.0 / BSD 3-Clause |

| Name & Version | Purpose in Project | License |
| --- | --- | --- |
| **Geopy** 2.5.0 | Geographic calculations in the Python codebase. | MIT |
| **python-dotenv** 1.2.4 | Loads local Python environment configuration. | BSD 3-Clause |
| **Orval / esbuild** 8.30.0 / 0.28.2 | Generates API client code and bundles server code. | MIT |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| OpenAI API | Generates a short recommendation summary from approved procurement history. | Sponsor-provided |
| Google Gemini API | Optionally generates supplier advice from stored illustrative offers. Deterministic eligibility and anomaly checks are performed separately. | Free |
| OSRM public routing service | Calculates driving distance and duration between coordinates. | Free |
| OpenStreetMap Nominatim geocoding service | Converts an address to coordinates. | Free |
| Google Fonts | Loads fonts used by the web UI. | Free |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit hosting and compute — Replit | Hosts the ProcureFlow web app and Express API. The web artifact is built and served as static files; the API runs as a Node service under /api. | Configuration targets autoscale. The workspace config does not disclose the Replit account's billing plan or deployment region. |

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit PostgreSQL module — Replit environment | The app's PostgreSQL database stores accounts, sessions, procurement records, decisions, invoices, and signature evidence. The workspace declares PostgreSQL 16 and connects through DATABASE_URL. | PostgreSQL 16 is specified. The database host, plan, and region are not identified in the project files. |
| Replit environment variables / Secrets — Replit | Supplies runtime configuration and credentials to the server-side services. | Provider is Replit; workspace billing plan and region are not shown. No secret values are included here. |
| Application authentication — team-built, running in Express | Handles signup, login, sessions, and logout. Password hashes and session records are stored in PostgreSQL. I found no external identity provider such as Auth0 or Clerk. | No separate provider plan or region. |
| OpenAI API via Replit AI integration — OpenAI / Replit integration | Generates recommendation summaries from approved procurement history. | Replit integration configuration is present; whether usage is free, paid, or sponsor-funded, and the processing region, are not available in the workspace files. |
| Gemini API — Google | Optional AI-generated supplier advice. It is called from the private Python advice service when configured and requested. | Google account billing/quota tier and processing region are not specified in the project files. |

| Service & Provider | Purpose in Project | Plan / Region |
| --- | --- | --- |
| Private Python advice process — Replit workflow | Runs the Python API on loopback port 8000; Express calls it for supplier checks and optional Gemini advice. | The project declares this as a private console workflow, not as a separate production deployment service. The checked-in production artifact configuration explicitly covers the web app and Express API; production availability for this Python process needs confirmation. |
| OSRM public routing — OSRM project | Routing helper code can request driving distances and durations. The current ProcureFlow offer flow uses illustrative distances instead; these helpers are not wired into that flow. | Public endpoint; no selected region or service plan is configured. |
| Nominatim — OpenStreetMap | A separate Python utility can geocode addresses to coordinates. It is not part of the current ProcureFlow UI request flow. | Public service; no selected region or paid plan is specified. |

## 4. AI Tools, Models & Datasets

### 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| GPT-5 mini (gpt-5-mini) | OpenAI | Pre-trained | Proprietary, provider-hosted model; use is governed by OpenAI API and applicable Replit integration terms, not an open-source model license. The project does not pin an immutable model snapshot. |
| Gemini model (GEMINI_MODEL) | Google Gemini API | Pre-trained | Proprietary, provider-hosted model; governed by Google's applicable API terms, not an open-source model license. The project does not pin an immutable model snapshot. |
| GPT Image 1 (gpt-image-1) | OpenAI | Pre-trained | Proprietary; OpenAI API and applicable Replit integration terms. |
| GPT Audio (gpt-audio) | OpenAI | Pre-trained | Proprietary; OpenAI API and applicable Replit integration terms. |

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| GPT-4o mini Transcribe (gpt-4o-mini-transcribe) | OpenAI | Pre-trained | Proprietary; OpenAI API and applicable Replit integration terms. |

### 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| **Account, session, and authentication records** | User signup and sign-in; stored in the app's PostgreSQL database. No public URL. | Private, self-collected | Private application data; no separate dataset license identified. | Y |
| **Procurement requests, approvals, and invoices** | User-entered app records stored in PostgreSQL. No public URL. | Private, self-collected | Private application data; no separate dataset license identified. | Y |
| **OpenAI recommendation inputs** | Derived from approved procurement records and sent to OpenAI through the server-side integration. | Private, self-collected; used for inference | No separate dataset license identified; provider API terms govern processing. | N |
| Gemini advice inputs | Derived from a stored request and its illustrative offers; sent to Google Gemini when advice is requested. | Private | No | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Illustrative supplier offers | artifacts/api-server/src/lib/procurement.ts. | Synthetic | No | N |
| Python synthetic supplier catalogue | src/database_w_input.py | Synthetic | No | N |
| Test and evaluation fixtures | Project test files, including synthetic procurement records and mocked provider responses. Tests use temporary databases; documentation says SQLite snapshots are opened read-only. | Synthetic and self-created | No | N |
| Historical SQLite catalogue snapshots | src/business_supplier_selection.db<br><br>backups/business_supplier_selection_backup.db; | Private | No | Y |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☐ Yes ☐ No

If yes, list them below:

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| **Replit Agent** | Project planning and codebase assistance. The project task history also indicates AI-assisted implementation and testing work. | Major |
| ChatGPT | Create backend code | Moderate |
| Gemini | Help in writing documentation | Moderate |
| | | |
| | | |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

| |
|---|
| AI-generated output was treated as a draft and checked against project requirements and existing architecture. Project records show type-checking and automated tests, including isolated database fixtures and mocked provider calls. My team remains responsible for deciding what to accept, handling data safely, and deploying the result. |

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

| **Sponsor** | **Resource (API / credits / platform / data)** | **How It Was Used in the Project** |
|---|---|---|
| Replit Agent | API | I did not call external AI, mapping, or other provider APIs. I used read-only workspace tools to inspect project files and configuration. |
| Replit Agent | Credits | This session may use Replit Agent credits, but I can't see the amount or whether they are free, paid, or sponsor-provided. |
| Replit Agent | Platform | Replit Agent in this Replit workspace. The underlying model and its version are not disclosed and may vary. |
| Replit Agent | Data | I inspected source code, configuration, documentation, and task summaries. I did not inspect database row contents or access secrets. |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

| |
|---|
| No, the sponsor resources were not strictly essential to the core business logic of the prototype. Although it help a lots in the development, they still can replace by self-hosted models or other AI models without affected the core procurement, database, and user interface functionalities. |

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

The app stores account names and emails, password hashes, session-token hashes, procurement requests, delivery details, requester/manager names and comments, approvals, invoices, and typed signature records in PostgreSQL. It also stores hashed identifiers for authentication rate limiting.
For AI features, approved purchase history is sent server-side to OpenAI for recommendations. Supplier-advice requests send offer details, prices, budget, deadline, and delivery estimates to Gemini when configured. Passwords are scrypt-hashed; session tokens are stored as hashes, and cookies use HttpOnly or SameSite protections and Secure in production. Same-origin checks, input validation, and server-only provider credentials are also used. Sessions last seven days. No defined retention period or automatic deletion for account and procurement records.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

The application faces limitations including potential AI inaccuracies, unverified input data, provider outages, single-provider dependencies, and unestablished production availability for its private Python service and Gemini model. To mitigate these risks, AI features remain strictly informational and non-authoritative, supported by deterministic checks such as eligibility and anomaly validation, response filtering for ineligible suppliers, UI disclosures, and technical safeguards like strict timeouts, error handling, and summary caching. Nonetheless, vulnerabilities persist due to the lack of alternate-provider fallbacks, unverified model availability, and the absence of independent sentence-level fact-checking.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☐ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

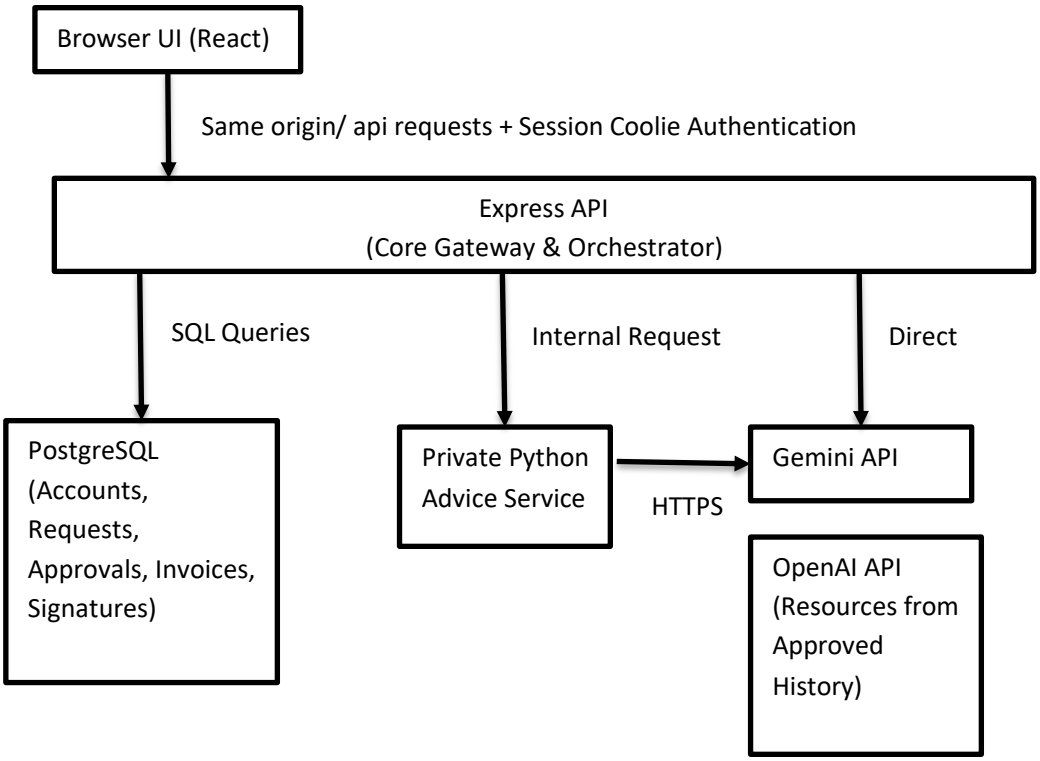
## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date] 25/10/2026

# Appendix


Browser UI (React)
Same origin/ api requests + Session Coolie Authentication
Express API
(Core Gateway & Orchestrator)
SQL Queries
Internal Request
Direct
PostgreSQL
(Accounts,
Requests,
Approvals, Invoices,
Signatures)
Private Python
Advice Service
HTTPS
Gemini API
OpenAI API
(Resources from
Approved
History)